A FRAMEWORK NOTE

Pricing Geopolitical Risk

What markets know about war, what they structurally cannot — and how to position

The index is calm almost everywhere, and almost nowhere is the index the right instrument. This note makes the case from a century of market history (Part I), maps where the risk actually lives (Parts III), turns it into trade structure (Part IV), catalogues the cases and the chokepoint thread (Part V), and sweeps the live 2026 board by region (Part VI).

Framework and risk disciplines only · not investment advice · 2026 figures move

PART I – II

The Argument, the Board & the Two Poles

What markets know about war, what they structurally cannot — and how to position

A reader's edition, in two parts. Part I is the argument, as narrative. Part II maps the board as it stands in the late summer of 2026 and turns the argument into positioning. It is the applied companion to a book about money and the nine years Europe spent not *going to war before 1914; it stands alone. Framework, not advice — the discipline is the product.*

PART I — The Argument

Here is the single most useful thing the historical record has to say to a trading desk, and it is counterintuitive enough that people rediscover it, expensively, every few years.

On the morning of 17 January 1991, the United States began bombing Iraq. Oil, which had nearly doubled since Saddam invaded Kuwait the previous August, fell thirty per cent — its largest one-day decline on record. The S&P rose about four per cent. The Bank of England, in its quarterly bulletin, wrote the explanation down in plain language: equities and bonds rose sharply "as news of early successes for the allies encouraged participants back into the market… the largest-ever one day decline in oil prices helped the rallies."

The war did not end that day. It had just started. What ended was the *uncertainty* — the fat tail the market had been carrying since August, in which Saddam torched the Gulf's oil fields and closed the Strait of Hormuz. Allied air supremacy from the first night took that tail off the table, and the premium built to price it evaporated in a session. Buy the invasion, sell the phony war.

That is not a quirk of 1991. It is the mechanism, and once you see it you see it everywhere.

Two different skills, routinely confused

A market does two very different things when a war looms, and it is good at exactly one of them.

The first is **anticipation**: pricing the probability that a war happens at all. This is forecasting a political decision that, in most cases, the decision-makers themselves have not yet made — and markets are bad at it. They cry wolf, because most scares fizzle. They price the wrong sign: in September 1939 the US market *rose* twenty-one per cent on the outbreak of the Second World War, because for a neutral arms supplier the war looked like a boom — and it did not crash until the fall of France, eight months later, turned a distant profit into a systemic threat. They get blindsided by surprises: Pearl Harbor drew a first-day drop of under four per cent, partly because a genuine surprise offers nothing to anticipate, partly because war risk was already half in the price. And in the summer of 1914, the canonical case, the money market said nothing at all until the Austrian ultimatum on 23 July — because the decision had been taken privately on the 5th and kept invisible until then. The market's first chance to price the crisis was the ultimatum, and the ultimatum was not a step toward the war. It was the war becoming visible.

The second skill is **resolution**: pricing the *consequences* of a decision once it has been taken. Here the market is fast, large, and correct — the oil premium collapsing the night Desert Storm began, the relief rally on the Munich accord, the market that recovered its whole loss the week of Inchon in 1950. This is the market's real talent: not prophecy, but arithmetic on a decision already made.

The practical upshot is a discipline: **do not trade the forecast; trade the resolution.** A calm market is not a forecast of peace. It is the market declining to handicap a decision no one has made — the precise error a generation of observers made in July 1914, and the precise error being made about Taiwan today.

The ladder: what kind of war is it?

Before you ask *how much*, ask *what kind* — because the market sorts wars not by size or bloodshed but by one criterion: could this go *general*? Could it become the systemic war that pulls in the great powers, at which point choosing a side stops mattering? That question produces a ladder with four rungs.

Localised wars, the market ignores. The Second Balkan War of 1913 — a real war — moved the great markets not at all. The Falklands barely dented sterling; the damage sat on Argentina. North Korean missile tests spike and fade. India and Pakistan can fight while global markets shrug. If a war is localised in its *impact*, the great markets discount it — and here their calm is roughly informative, because it genuinely stays contained.

General but judged unlikely, the market discounts. The Berlin Airlift of 1948–49 was a prolonged US–USSR standoff, and the American market *rose twenty-one per cent through the war scare* and shrugged at the blockade. It gave no relief rally when the blockade lifted — because the war had never been in the price. (That missing rally is a clean diagnostic: resolution rallies only where the war was actually priced.)

General and survivable, the market prices — dynamically. Korea in 1950 is the cleanest case. It broke thirteen per cent on the June invasion, feared that summer as the opening of a third world war; recovered the entire loss once Inchon made the war look contained; broke six per cent again when China's intervention revived the general-war fear; and made new highs once it settled into a limited war. The *same war* repriced up and down as its probability of going general rose and fell. This is where oil wars live too, and it is the rung where the resolution trade works.

Existential, the market cannot price at all. In October 1962, through the most dangerous fortnight in human history, the S&P fell a few per cent — and its low came *before* the most dangerous days. It rose on the morning the blockade of Cuba was enforced. The reason is not that the market judged the risk small. It is that there is no version of the world in which the bearish position pays off: a claim on a future that, in the catastrophic branch, does not exist. **You cannot price a catastrophe you will not survive to collect on.** The discrimination test has a ceiling, and existential risk sits above it, unpriced not because it is improbable but because the instrument cannot represent it.

Keep that ceiling in mind. It is the most important thing on the board for anyone thinking about Taiwan.

Read the exposed instrument, not the index

The second discipline: the signal does not live in the index. It lives in whichever instrument is directly exposed, and that instrument is different in every war.

In the Gulf and over Iran it is **oil**. In 2022 it was **energy** — the S&P actually closed *up* on the day Russia invaded Ukraine, and the VIX peaked at a mere 36, because for Western equity the war was an energy shock, not a systemic threat; the war went into Brent (up thirty-four per cent) and into the ruble, not the S&P. In 1905 it was a **sovereign bond** — French savers watched their Russian holdings halve while the French government's own paper sat still, a portfolio crisis invisible to any index. At Suez in 1956 it was **reserves** — under a fixed exchange rate the pressure showed not in a price but in Britain's draining gold and the denial of IMF access, which is what forced the withdrawal.

A generalist watching the S&P through any of these would have seen almost nothing until it was over. The person watching the exposed instrument saw the whole thing. The corollary for the next war: **identify the instrument before the crisis, not during it.**

Iran: the tail you can actually trade

Iran is the clean, live example, because the exposed instrument — oil — is liquid, and the tail — the Strait of Hormuz — is survivable and collectable, so the market *can* price it.

Watch it across two episodes. In June 2025, Israel and then the United States struck Iran; oil spiked about ten per cent on the fear Hormuz would close. Then Iran's response was telegraphed and symbolic, the strait stayed open, and oil fell *below* where it had started, while equities made new highs — the tail priced, then removed, then unwound, in days. Then in the spring of 2026 the tail actually *materialised*: Hormuz was shut, roughly eight million barrels a day came off the market, and oil ran from the high fifties to a hundred and fourteen dollars and *held above a hundred for two months*. Same instrument, two completely different behaviours — because the first was a scare and the second a real closure, and the size and, crucially, the *duration* of the move discriminated between them.

That is the tradeable structure: the Hormuz premium is real only while the strait is actually shut; it collapses the moment the closure is removed; and the market is poor at forecasting whether the closure comes but fast at pricing it once it does. A calm oil price is not a forecast that nothing happens — it is the market declining to handicap Tehran's undecided choice.

And there is now a new instrument pointed at exactly the channel markets are worst at. Prediction markets price the *decision itself*: as of this writing the crowd puts US forces entering Iran near ninety per cent and normal Hormuz traffic by September under half, with tens of millions staked. For the first time you can *decompose the war premium* — probability off the prediction market, size off the oil curve — and in the spring of 2026 the reopening odds moved first and the oil price followed. Treat that instrument with the suspicion you would any thin, manipulable market: it can cry wolf, and the resolution still dominates. But it is genuine new information in the one place the market has always been blind.

Taiwan: the tail you cannot

Taiwan is important precisely because the intuitions everyone carries about it are wrong.

A war over Taiwan is a candidate *general* war — US against China, possibly nuclear — which puts it near the Cuba end of the ladder, not the Gulf. So two things follow: the market will be poor at pricing it in advance, because it is a decision Beijing has not made; and to the degree it approaches the existential, it drifts toward *un-priceable*. Both mean the same warning: **calm markets are not evidence a Taiwan war is unlikely.** They are the July-1914 signature.

The data fits exactly. When Nancy Pelosi visited Taiwan in August 2022 and China answered with its largest-ever blockade drills — missiles over the island, a rehearsal of quarantine — the markets barely moved. Semiconductors, the exposed instrument, took the biggest hit at minus three and a half per cent; the S&P fell under one per cent; everything was *higher* within a fortnight. The market discounted that episode as a drill, and was right that time. But a three-and-a-half-per-cent semis wobble that reverses in two weeks tells you the market judged *that episode* would stay below the general threshold. It tells you nothing about a priced invasion. If Beijing decided, the repricing would come at the resolution, be violent, and land in chips and Chinese assets first — not in the ambient calm beforehand.

Note the instrument. For Taiwan it is **semiconductors** — the island fabricates the overwhelming majority of the world's advanced chips, so a Taiwan war is a *silicon* shock the way a Gulf war is an *oil* shock. Watch the semiconductor complex, the Taiwan dollar, the yuan; the S&P is the last place the signal shows. And treat the direct-war tail the way you treat any existential risk: it is not a spot premium you can harvest, it is a convexity question, because the market structurally cannot put it in the price. The absence of a Taiwan premium is not the absence of the danger. It is the instrument hitting its ceiling.

Finance as a weapon — and why the weapon is decaying

There is one more move on the modern board the pre-1914 world only hinted at: finance deployed not as a barometer but as an outright weapon. The lineage is three freezes. In 1941 the United States froze Japanese assets and embargoed oil, and cornered Japan toward war. In 1956 it denied Britain the sterling and IMF support it needed, and forced a great power to abort a war it was winning. In 2022 the G7 froze roughly three hundred billion dollars of Russian central-bank reserves — and it punished without halting. One weapon, three outcomes: provoked, coerced, punished. The effect is documented in every case; the *decisiveness* varies and must never be assumed.

The important dynamic is what 2022 set off, because its most important audience was Beijing. China holds a vast dollar reserve and, until recently, an enormous Treasury position — and it watched a peer's reserves immobilised overnight. It has been hardening ever since: the People's Bank has been buying gold for twenty consecutive months, to some 2,346 tonnes, and kept buying straight through a sixteen-per-cent price slide — the signature of structural rebalancing, not a trade. Gold is still under ten per cent of its reserves against roughly seventy for Western central banks, so there is a great deal of room left. Its Treasury holdings have fallen to around six hundred and ninety billion dollars, the lowest since 2008 and half the 2013 peak, and it is building payment rails that route around the Western messaging system.

The logic is precise: **a Treasury bond is a hostage; a bar of gold in your own vault is not.** Gold is the anti-freeze asset — a claim on no one, held at home, settled without permission. So the gold pile and the Treasury run-off are one move — sell the hostage, buy what cannot be seized — and the implication is directional and dated: **the coercive value of the US financial weapon over China is a wasting asset.** Germany, squeezed at Agadir in 1911, spent the next three years drawing gold into the vault and reforming bank liquidity — a reform completed the very month the war began, by which point the squeeze could not have been repeated. China is doing the same reform *ahead* of its crisis.

Two disciplines keep this from being a story you lose money on. Do not overclaim the hostage is freed: six hundred and ninety billion is still an enormous exposure, gold cannot pay for imports at wartime scale, and the rails still lean partly on Western messaging. China has *reduced* the weapon's bite, not neutralised it. And — the deepest point — **the gold documents capability, not a decision.** Prudential reserve-building and preparation for aggression are identical on a balance sheet. Reading the gold as proof of war plans is the field's deepest error: reading exposure as intent, capability as decision.

PART II — The Board Today, and How to Position

The thesis in three sentences

Markets are poor at forecasting *whether* a war happens and excellent at pricing its *consequences* once the decision is taken; the gap is where the mistakes and the trades live. They price the probability of a *general* war — not war as such — so they ignore localized wars, price survivable ones (and mean-revert when the tail is resolved), and fall silent before the existential ones they cannot represent. And finance is now itself a weapon whose coercive value **decays as targets harden** — the most important slow fact on the board, and it is being made in Beijing.

The board

THEATER	RUNG	EXPOSED INSTRUMENT	STATE (MID-2026)
Iran / Hormuz	general, *survivable* → priceable	**Oil**	Live, unresolved. Hormuz shut in spring 2026 (oil +52%, held >$100 two months); fragile, violated ceasefire; oil oscillating ~$75–82. The tradeable tail.
China / Taiwan	candidate *general/existential*	**Semiconductors**, CNH/TWD	Calm — and calm here is *not* informative. Pelosi-2022 drills moved semis only −3.7%, reversed in a fortnight. The un-priceable tail.
The sanctions weapon	—	Reserves, gold, payment rails	The 2022 freeze set off structural de-risking; its value is decaying. The biggest slow fact.
Russia / NATO	escalation tail → un-priceable	European **energy**; the freeze	Weapon deployed; direct-war tail near the ceiling — silence ≠ safety.
Norway (the anti-China)	—	Gas, energy equities (*not* NOK)	Europe's gas lifeline; the ~$2.2tn fund is the *pro-Western-system* mirror of China's exit.
NK, India–Pakistan	localized	regional FX/equity	Great markets ignore; here calm *is* roughly informative.
Climate / AI tails	above the ceiling	—	Un-priceable; market silence is not reassurance.

China and Norway — the two poles of sovereign money

The sharpest pairing on the board. Two resource-funded state pools of extraordinary scale, de-risking in **opposite directions** — the same "sovereign money as a geopolitical actor" phenomenon pointed both ways at once.

China exits the Western system: opaque, sanction-*proofing*, ~2,346 tonnes of gold and its own payment rails, Treasuries down to ~$690bn. Building a fortress against the weapon. The forward read: US financial leverage over China erodes every year it de-risks — but the gold documents *capability, not a decision*, and the exposure is still enormous, so the weapon is *reduced*, not neutralised.

Norway anchors that same system. Its sovereign fund — **~$2.2 trillion, the world's largest, owning about 1.5 per cent of every listed company on earth** — is transparent, rules-based, and sanction-*enforcing* (it divested Russia in 2022), and, under a fiscal rule that reinvests all petroleum revenue and spends only the expected real return, a structural, price-insensitive *buyer* of global equities. It is not a fortress; it is a load-bearing wall of the side that wields the weapon, and the stabilizing counterweight to the gold/de-dollarization bid.

The krone trap, because it is a live way to lose money. People do trade NOK as a petrocurrency — it sits in the commodity-FX bucket with the Canadian and Australian dollars and had a positive oil beta historically. But that beta broke down: weakening after the 2015 oil crash and *spectacularly* in 2022–23, when Norway became Europe's largest gas supplier with record energy terms-of-trade and the krone nonetheless fell to multi-decade lows (USD/NOK roughly 8.3 in 2021 to 10–11 in 2023–25). The naive "oil-currency, long the energy shock" trade failed — because NOK is the least-liquid G10 currency and trades as a *risk-sentiment proxy* first (sold in any risk-off regardless of energy), with rate differentials, a transition-risk premium on fading future petroleum revenue, and the sovereign fund's own FX flows all pulling the other way. The lesson stands, sharpened: **NOK is a risk proxy first and an energy proxy a distant second** — the exposed instrument (gas) went the right way while the obvious currency proxy was driven by something else. *Read the instrument, not the naive proxy.*

And this is not a modern curiosity — it is the same lesson the pre-war material turns on, in two forms. In 1905, France's money market read *calm* not because France was safe but because its risk sat in a different instrument entirely — Russian bonds in French portfolios; the obvious instrument showed a null that wasn't one. At Agadir in 1911, Berlin's discount rate barely registered the squeeze because a mechanical factor, the quarter-end settlement, swamped it — the crisis was in the quantities, not the price; the obvious instrument was loud with the wrong signal. NOK is a live instance of both: silent on energy because the strength sits elsewhere, and swamped by liquidity and rates besides. The obvious instrument can mislead by nullity or by contamination — which is why you read the one that actually carries the exposure.

The one line to keep

The risks that would end the game are the ones the market cannot price at all — so on those, its silence is not a signal. Everywhere else: don't trade the forecast, trade the resolution, in the instrument that is actually exposed, and remember the financial weapon is a wasting asset. Somebody still has to be in the room to ask the money question. The market cannot ask it for you.

Sources are reproducible or cited: daily equity data from the S&P 500 (back to the 1920s) and oil from the WTI and Brent series, via public feeds; the 2003 war-risk factor from Sack and Rigobon's The Effects of War Risk on U.S. Financial Markets *(2005); the 1991 attribution from the Bank of England Quarterly Bulletin; reserve, gold, fund, and holdings figures from official sources (US Treasury TIC, the People's Bank of China, Norges Bank). The 2026 figures are contemporary and move; check the dates before relying on them.*

PART III

The Instrument Problem — at a Glance

One idea: the index is calm almost everywhere, and almost nowhere is the index the right instrument.

A framework and its risk disciplines, not a recommendation sheet. It tells you where the risk lives and what shape the trade takes — never when. Most of these are convex, negative-carry, timing-hard, and several are crowded.

The idea in a paragraph

When a risk builds, the signal shows up in whichever instrument is directly *exposed* — not in the broad index, and often not in the "obvious" proxy. Read the wrong instrument and it misleads you in one of three ways: it shows a **null that isn't** (the exposure sits elsewhere), it's **contaminated** (a mechanical or structural factor dominates the price you're watching), or it's **un-priceable** (no instrument can carry it, so silence is not information). This isn't a new observation — it's the oldest lesson in market history. France's money market read calm in 1905 while the risk sat in Russian bonds in French portfolios (*the null that isn't*); Berlin's discount rate barely moved in the 1911 Agadir squeeze because the quarter-end swamped it, and the crisis lived in the *quantities* (*contaminated*). The edge today is the same: **the crowd watches the calm index; you find the instrument that actually carries the exposure.**

Three failure modes → three trade shapes

FAILURE MODE	WHAT IT LOOKS LIKE	TRADE SHAPE
Null that isn't	Obvious instrument silent; risk sits elsewhere	Buy cheap convexity in the *exposed* instrument, where the tail is under-priced
Contaminated	A mechanical/structural factor dominates the naive proxy	Express in the instrument that *tracks*; trade the *divergence* proxy-vs-real
Un-priceable	No instrument can carry it	Insurance or avoidance — never premium harvest

And one timing rule across all three: **trade the resolution, not the forecast** — don't bleed theta betting on *whether*; structure to get paid on the *repricing at the decision*.

The live map (mid-2026)

SITUATION	OBVIOUS INSTRUMENT (MISLEADS)	WHERE THE RISK ACTUALLY LIVES	TRADE SHAPE	WHAT KILLS IT
China / Taiwan	S&P / China index — *calm; null*	Semis (SOX/TSM), CNH, TWD; an un-priceable tail	Convex insurance: OTM put spreads on SOX/TSM/EWT, USDCNH topside calls, long TWD vol	Bleeds for years; in a *real* war the hedge may not settle — also a supply-chain / avoidance problem
Dollar-weaponization / de-dollarization	DXY — "dollar still king"; *null*	Gold, China's TIC Treasury line, CIPS, RMB settlement share	Long gold as an *allocation* to the central-bank bid; miners for beta	Crowded and extended; a real-rate spike or dollar squeeze
Norway / NOK	The krone as an energy proxy; *contaminated*	Gas (TTF), terms of trade, energy equities	Energy view via Equinor / energy equities / TTF — **not** the currency; NOK is a risk/rates trade	Conflating the two — trading NOK expecting an energy beta
Russia / Ukraine	Western indices (calm) and the ruble (managed); *null* + *contaminated*	European gas (TTF), wheat, the Urals–Brent discount, European industrials	Long TTF / wheat escalation optionality; the Urals discount via flows/tanker rates	TTF and wheat mean-revert hard; negative carry; **don't trade the ruble**
Iran / Hormuz	The index; *null* (the *tradeable* tail)	Oil (Brent/WTI), VLCC tanker rates, Hormuz prediction-market odds	Brent call spreads as insurance when the strait is open and vol is cheap; sell scare-richness; RV: crowd-odds vs oil-implied	Theta bleed through calm; a *real* closure if you're short
US fiscal	S&P at highs; *null*	Long-end / term premium, MOVE, gold, auction demand	5s30s steepeners; long bond vol / payer swaptions; gold as the no-anchor hedge	Long end rallies in risk-off; slow; negative carry — has gored many
AI / concentration	Index level (new highs); *contaminated*	Breadth; the semis complex; datacenter/power + the private credit funding it	Breadth (equal-weight RSP vs SPY); dispersion (long single-name vol / short index vol)	Bleeds in a momentum melt-up; shorting concentration is a graveyard
Private credit / CRE	S&P / bank-equity indices; *null*	Private-credit marks, CRE, regional-bank deposits / HTM losses	KRE puts, CMBX, BDC credit	The catalyst is a *forced mark* — patient, negative carry until one arrives
Japan / JGBs	The yen; *contaminated*	The JGB long end; the carry-unwind spillover	Short JGBs / long payer optionality; long yen vol for the global spillover	The widowmaker — the BOJ can pin it; carry

Cross-cutting tell: **gold rising *with* real rates** is the single best signal that the market's attention has moved to a non-obvious instrument (the central-bank sanction-proofing bid, orthogonal to yields). When the textbook correlation breaks, the signal has migrated.

The disciplines that save money

1. **Trade the resolution, not the forecast.** Structure to get paid at the decision, not to bleed on *whether*.
2. **Buy the *exposed-instrument* vol, not the index vol.** Oil vol for Iran, semis skew for Taiwan — usually cheaper and cleaner than SPX.
3. **Un-priceable tails are insurance, not carry.** Size tiny, expect bleed, and know the hedge may not pay in the real tail.
4. **Beware the crowded, negative-carry consensus.** Long gold, short JGBs, short-duration-fiscal, short-concentration are all "everyone knows." The edge is strongest where the crowd is still watching the *wrong, calm* instrument — not where the right trade is already consensus and expensively carried.
5. **Prefer the relative-value version.** Where a naive proxy *and* a real instrument both exist, the *divergence* is often the cleanest trade: crowd-odds vs oil-implied, gold vs real rates, RSP vs SPY, the Urals discount.

6. **Size for convexity and patience.** Small, defined-risk, and *roll* — not levered directional bets on timing.

The honest caveat

Knowing *where* the risk lives is necessary, not sufficient: most of these are right on **instrument** and hard on **timing**, and several are crowded. The framework's real gift is **negative** — it stops you expressing the view in the wrong instrument (long NOK for gas, short SPX for Taiwan, trading the ruble for Russian stress), which is where most people quietly lose. Everywhere else, the one line:

> ***The index is calm almost everywhere. In almost none of these is the index the right instrument.*** *Find the one that carries the exposure — and treat every calm headline print as the July-1914 tape until you have.*

Companion to a longer piece, Pricing Geopolitical Risk*, and to a book on money and the pre-1914 crises where this "instrument problem" is the recurring finding — from the* fonds russes *in 1905 to the krone in 2026. Framework and risk disciplines only; not investment advice; the 2026 figures move.*

PART IV

Trade Expressions

Concrete structures for each situation on the live map: the thesis, the primary expression, a relative-value or alternative, the carry/sizing note, and what kills it. These are illustrative of how the framework maps to expressions — not investment advice, not a solicitation. Almost all are right on instrument and hard on timing; several are crowded and negative-carry. Size as risk discipline dictates.

How to structure anything on this list (the meta-rules)

- **Buy the *exposed-instrument* vol, not the index vol** — oil vol for Iran, semis skew for Taiwan. It's usually cheaper and cleaner than SPX, and it's the thing that actually moves.
- **Prefer defined-risk convex structures** (spreads, not naked shorts/longs) — given negative carry and timing uncertainty, you want bounded bleed and bounded loss.
- **Trade the resolution:** own optionality *into* cheap vol / calm, and **monetize or roll on the spike** — don't hold to expiry hoping the forecast comes true.
- **Prefer the relative-value version** where a naive proxy *and* a real instrument both exist — the *divergence* is often the cleanest, best-carry trade.
- **Size by role:** tail hedges as *insurance* (bps of NAV, expect bleed); structural themes as *allocations* (scale in, don't chase); RV as *risk-budgeted* pairs.

Trading the quiet — the no-catalyst book

The hard question the framework provokes: if you *trade the resolution, not the forecast*, what do you do when there is no resolution — which is **most of the time**? "No catalyst" is the base rate, not the exception. It is the subject of the book this accompanies: nine years in which Europe kept *not* going to war. You do not sit in cash waiting for one. You run a book that pays in the quiet and carries a cheap overlay for the rare decision.

1. Don't fight the calm. A quiet market is not a mispricing — it is the market correctly declining to price a decision no one has made. So don't manufacture a catalyst or force a forecast where you have no edge. Most of the time the right geopolitical-tail position is *small and cheap*, and the P&L comes from elsewhere. Perma-bear-on-a-war-that's-years-away is how you bleed to death.

2. Run the engines that need no catalyst (the base book). - Structural / carry — gold as the central-bank-bid allocation, the de-dollarization plumbing, AI-breadth. Secular flows; they grind regardless. - **Relative value / divergence** — gold vs real rates, Equinor vs NOK, RSP vs SPY, the Urals discount. Persistent mispricings that mean-revert on their own clock, catalyst-free.

This is the "peacetime money market" of the book — the part that compounds through the nine years.

3. In the quiet, be an insurance *seller* — of scares. "No catalyst" is exactly what makes selling scare-premium pay, because *most scares fizzle*. So in calm periods you flip from buyer to **seller of the phony-war premium**: fade the rich vol/upside that spikes on a *localized* flare-up the market was always going to ignore. The discrimination ladder is what makes it safe — **sell the localized scares (the great markets ignore them anyway); flip to buyer or flat the moment one crosses toward the *general-war* rung.** The one trade you cannot get wrong is selling the scare that's actually general — the July-1914 mistake.

4. Accumulate the cheap tail *because* there's no catalyst. The calm is when exposed-instrument convexity is cheapest — nobody wants Brent calls when the strait is open, or SOX tail when Taiwan is quiet. So the quiet is the **accumulation phase** — buy straw hats in winter — sized as insurance you *expect to bleed on*, and financed partly by the scare-selling in (3): sell the rich front, own the cheap back (calendar/ratio structures). Holding cheap convexity *through the boredom* is the whole edge, and the reason it stays cheap is that almost no one else can.

The no-catalyst book in one line: carry + RV + scare-selling pay the bills and compound through the quiet; a small, cheap, always-on tail overlay (bought when cheapest, financed by the scares) costs a little and pays rarely and hugely; and the framework's job in the calm is to prevent the two ruinous errors — **forcing a war bet with no edge, and selling the one scare that's real.**

Finding the mispricing (before you express it)

A worked detector for *where* the edge is: decompose the exposed instrument's premium into `P(disruption) x size`, read **P** off the prediction market (Kalshi) and **size** off the instrument's vol/curve, and trade the **gap** (odds-rich + premium-cheap -> buy convexity; premium-rich + odds-cheap -> fade). Full worked example (Hormuz, two-sided, with the live OVX~55 reading) in `mispricing-worked-example.md`; the systematic long-only version is the **geopolitical sleeve** in the equity stock-picker (`pari_mutuel_geopolitical_sleeve.md`).

The expressions

1. China / Taiwan — the un-priceable tail (insurance, not harvest)

- **Thesis:** calm index ≠ low risk; the exposure is chips and the yuan, and the full-war tail can't be priced — so buy cheap convexity where it's exposed.
- **Primary:** 6–18m OTM **put spreads on SOX or TSM** (buy ~15–25% OTM, sell a further-OTM put to cheapen). **USDCNH topside call spreads** — CNH is the pressure valve, and FX convexity is often cheaper than equity puts. Long **TWD vol / USDTWD calls**.
- **RV:** long **semis skew vs SPX skew** (semis tail is cheap relative to its systemic importance); or SOX puts financed by SPX puts — own the *exposed* tail against the *index*.
- **Carry/size:** bleeds for years; size as insurance and roll.
- **Kills it:** in a *real* war, exchange/settlement/broker risk means the hedge may not pay — so pair with real-economy **supply-chain diversification** (avoidance, not just options). China policy support can also cap CNH.

2. Dollar-weaponization / de-dollarization — structural gold

- **Thesis:** the signal is central-bank gold demand and the flows (TIC, CIPS), not DXY; gold rising *with* real rates is the tell.
- **Primary:** a **strategic gold allocation** (physical/ETF) as exposure to the price-insensitive central-bank bid; **miners (GDX)** for beta.
- **RV:** long **gold vs long-end real rates** (the divergence *is* the thesis) — a cleaner risk profile than outright gold.
- **Carry/size:** allocation, not momentum; extended and consensus, so scale in on dips, watch positioning.
- **Kills it:** a real-rate spike, a dollar-funding squeeze, a positioning washout.

3. Norway / NOK — separate the two views

- **Thesis:** NOK is a risk/liquidity proxy first, energy a distant second; don't express energy through the currency.
- **Primary (energy view):** long **Equinor / European nat-gas (TTF) / energy equities** — *not* NOK.
- **NOK view (if any):** treat it as a **risk-off / carry** expression, not energy.
- **RV:** long **Equinor vs short NOK** — own the energy, neutralize the currency's non-energy drift (know both legs: this isolates the petro-earnings from the risk-proxy beta).

- **Kills it:** conflating the two — trading NOK expecting an energy beta.

4. Russia / Ukraine — the exposed instruments, never the ruble

- **Thesis:** the war's economic signal lives in gas, grain, and the Urals discount; the ruble is a managed propaganda price.
- **Primary:** long **TTF (European gas) call spreads** for supply-disruption/escalation; long **wheat calls** (Black Sea).
- **Adjacent:** the **Urals–Brent discount** via **tanker rates (VLCC/Aframax) and shipping equities** — the sanctions-evasion gauge; **European industrials/utilities** as the gas-exposed real economy, hedged with TTF.
- **Avoid:** the **ruble** — capital controls, convertibility and settlement risk; not a signal, not a trade.
- **Carry/size:** TTF and wheat mean-revert hard → defined-risk spreads, monetize spikes.
- **Kills it:** mean reversion, a ceasefire, negative carry.

5. Iran / Hormuz — the tradeable tail (resolution trade)

- **Thesis:** the Hormuz premium is real only while the strait is actually shut; the market is poor at forecasting the closure and fast at pricing it.
- **Primary:** own **Brent call spreads** (~3–6m, 10–25% OTM) as Hormuz insurance **when the strait is open and oil vol is cheap**; monetize/roll when a scare spikes vol.
- **Fade the scare:** after a spike likely to be a *scare* (telegraphed/symbolic response), **sell rich upside/vol — as a spread, never naked** against a real closure.
- **RV / decompose:** compare **Hormuz prediction-market P(closure) to the oil-implied premium (size)** and trade the gap; **VLCC rates** as confirmation.
- **Carry/size:** theta bleed through calm → spreads, roll; the resolution is the payoff.
- **Kills it:** bleed through calm; a real, executed closure if you're short.

6. US fiscal — the long end, not equities

- **Thesis:** debt-sustainability risk shows in the term premium and gold, never in an equity index at highs.
- **Primary: 5s30s steepeners**; **long-end payer swaptions / TLT puts** for the fiscal tail; **gold** as the no-anchor hedge.
- **RV:** 30y **real yields vs gold**; **nominal 30y vs breakevens** (term premium or inflation?).
- **Carry/size:** negative carry, slow, whipsaw-prone (the long end rallies in risk-off) → convex, patient, small, defined-risk.
- **Kills it:** a risk-off long-end rally; fiscal consolidation; carry. (This trade has gored many — respect it.)

7. AI / concentration — trade the concentration, not the level

- **Thesis:** the index level hides that a handful of names carry the risk.
- **Primary: equal-weight vs cap-weight (RSP/SPY)** for breadth; **dispersion** (long single-name vol / short index vol) to monetize concentration.
- **Adjacent: semis** are the exposed instrument for both the AI upside and the Taiwan tail (asymmetric); watch the **private-credit / HY funding** of the datacenter-and-power capex — short it if the funding cracks.
- **Carry/size:** breadth and dispersion bleed in a melt-up → risk-budgeted RV, not outright short.
- **Kills it:** a momentum melt-up; dispersion staying pinned low.

8. Private credit / CRE / regional banks — where the marks don't move

- **Thesis:** stress hides in illiquid private marks; the public price is silent until a forced mark.
- **Primary: KRE puts** (regional banks); **CMBX** shorts on the riskier tranches (CRE); **BDC** equity/credit shorts for private-credit mark risk.
- **Carry/size:** negative carry until a catalyst (a forced mark / redemption wave); size for patience; mind basis risk in the proxies.

- **Kills it:** extend-and-pretend continues; rate cuts relieve CRE; no forced mark arrives.

9. Japan / JGBs — the spillover more than the bond

- **Thesis:** BOJ-normalization risk shows in the JGB long end and, more importantly, in what a carry-unwind does globally — not in USDJPY spot.
- **Primary:** short **JGBs / long JGB payer swaptions**; **long yen vol** (or long yen) to hedge the global carry-unwind spillover.
- **Carry/size:** the widowmaker — negative carry, the BOJ can pin it; small, convex.
- **Kills it:** BOJ yield control persists; carry.

The one-paragraph summary for the desk

Own convexity in the **exposed** instrument, not the index: **Brent** for Iran, **SOX/CNH** for Taiwan, **TTF/wheat** for Russia, the **long end/gold** for US fiscal, **breadth/dispersion** for AI, **KRE/CMBX** for credit, **JGB payers** for Japan — and **gold** as the structural read on the sanctions-weapon decay. Express energy as **gas and equities**, never the **krone** or the **ruble**. Trade the **resolution**, not the forecast; prefer **defined-risk spreads** and **relative-value divergences**; treat the **un-priceable** tails as insurance you expect to bleed on. The framework's edge is knowing *which instrument* — the timing and the sizing are still yours, and the honest part is that most of these are right on instrument and hard on when.

Companion to `instrument-problem-desk-note.md` and `pricing-geopolitical-risk-READER.md`. Framework and risk disciplines only; not investment advice; illustrative structures; 2026 levels move.

PART V

More Cases & the Chokepoint Thread

Every situation is a (rung, instrument) pair: ask "could it go general?" (localized → ignored / general-survivable → priced / existential → un-priceable) and "which instrument is exposed?" (never the index). This catalogue adds the historical cases worth running, draws out the chokepoint thread that ties them to the present, and then leans into the situations live today. Charted figures are reproducible (Yahoo/FRED); 2026 figures are cited and move.

A. Historical cases worth adding (each adds a *new mechanism*)

Six-Day War, June 1967 — localized, fast, relief rally; and the chokepoint that lingered. The S&P sagged ~5% as the crisis built and the war opened (92.7 → 88.4 on 5 June), then **recovered in full within a week** as Israel's decisive victory resolved it (back to 92 by 12 June). Buy-the-invasion, again — a fast, localized, US-uninvolved war the great markets shrugged off. But the *casus belli* was a **chokepoint** (Egypt's closure of the Straits of Tiran), and the lasting economic mark was another: the **Suez Canal closed for eight years** (1967–75). Lesson: the war is priced in a week; the *chokepoint* is priced for a decade.

Yom Kippur War, October 1973 — the oil *weapon*. The damage came not from the fighting but from the **OPEC embargo**: crude roughly **quadrupled** (~$3 → ~$12), and the West got stagflation. The purest "commodity-as-weapon" case, and the ancestor of the whole energy-instrument lineage — older than the 1941/1956/2022 financial freezes, and the reason "finance/commodity as weapon" is a category at all.

Iran–Iraq "Tanker War," 1984–88 — the *original* Hormuz. Attacks on Gulf shipping, Kuwaiti reflagging, US Navy escorts in the strait. This is the literal precedent for the Iran/Hormuz trade today — the base rate for how a Hormuz threat prices (oil + tanker rates + war-risk insurance), and a reminder that a chokepoint can be *contested for years* without being fully closed.

9/11, September 2001 — the surprise, and the infrastructure casualty. The exchange **closed for four days**; on reopening the S&P fell to a −11.6% trough within the week (1092 → 966), then recovered above its pre-attack level within a month. The Pearl Harbor pattern (a true surprise offers nothing to anticipate), with the added lesson that **the market plumbing itself can be the casualty** — the modern echo of the 1914 London close.

Brexit, June 2016 — the instrument problem in a political shock. The risk lived in **sterling** (1.46 → 1.29, ~−13%), *not* UK equities — the FTSE 100 *rose*, being full of global earners helped by a weak pound. And it was a **referendum**, so a clean anticipation failure: the market priced Remain right into the close, then gapped. Read the exposed instrument (the currency), not the index.

B. The chokepoint thread — one instrument-family across the century

Four of these cases are the same trade wearing different geography, and it runs straight to now:

CHOKEPOINT	CRISIS	ERA
Straits of Tiran	Six-Day War (the trigger)	1967
Suez Canal	Closed 8 years after 1967; the Houthi era again today	1967–75 · 2024–
Strait of Hormuz	Iran–Iraq Tanker War; Iran 2025–26	1984–88 · 2025–
Bab-el-Mandeb / Red Sea	Houthi attacks	2024– (live)

A chokepoint crisis does **not** price in the equity index. Its instruments are **oil/LNG routing, tanker and container freight rates, shipping equities, and war-risk insurance premiums** — and the resolution is binary and datable (the strait opens or it doesn't). It is the cleanest, most repeatable version of "read the exposed instrument," and there is a live one on the board right now.

PART VI

The Live Board by Region — 2026

The framework swept across the map. Every situation is a (rung, instrument) pair — could it go general (localized → ignored / general-survivable → priced / existential → un-priceable), and which instrument is exposed (never the index). Widening the aperture surfaces two cross-cutting instrument-families that dominate the current board: chokepoints and critical minerals. Live 2026 figures are cited and move; verify at use.

Red Sea & the Middle East — the chokepoint belt

- **Red Sea / Bab-el-Mandeb (live).** Houthi attacks continue; most carriers still route via the **Cape** (+10–14 days); Asia–Europe container rates **25–40%** above pre-crisis; a strike on Saudi tankers pushed **Brent back to $100**. Instrument: **container freight (SCFI/Drewry), shipping equities (Maersk, Hapag, ZIM), oil/LNG routing, war-risk insurance**. Two-sided: a durable reopening collapses freight ~25%; escalation spikes it. Localized disruption, priced through *routing* — not a systemic-war premium.
- **Hormuz / Iran (live, unresolved).** Fragile ceasefire; the tail is a strait closure — *survivable and collectable, so priceable*. Instrument: **oil (Brent/WTI), VLCC rates, Hormuz prediction-market odds**. (Full treatment in the Iran docs.)
- **Suez / Egypt.** The chokepoint's fiscal shadow — lost canal revenue stresses Egypt. Instrument: **EGP, Egyptian sovereign bonds** — not a global index.
- **East Med gas** (Israel/Cyprus/Lebanon/Turkey). Fields and pipelines; instrument: regional gas, specific E&P names.

Asia — semis, memory, and the mineral chokehold

- **China / Taiwan.** Candidate general/existential → the index is the *wrong, calm* instrument. Exposed: **semiconductors (SOX/TSM), CNH, TWD**. Un-priceable tail → insurance, not premium.
- **South China Sea / Philippines.** China–Philippines flashpoints (Scarborough, Second Thomas Shoal). *Localized with a general-war tail* via the US–Philippines treaty. Instrument: regional equity/FX and **shipping lanes**; the tail lives in the same place Taiwan does.
- **North Korea.** Regional, tail = Chinese involvement. Instrument: **KRW, Korean equities, and memory chips (Samsung/SK Hynix)** — the DRAM silicon-shock, Asia's second chip chokepoint.
- **India.** The border with China (Himalayas) is a *localized* risk; the bigger story is **structural** — India as the China+1 beneficiary. Instrument: Indian equities/rupee as an *allocation* theme, not an event trade.
- **Japan / JGBs.** BOJ normalization → the JGB long end and the **carry-unwind spillover**, not spot USDJPY.
- **The mineral chokehold (the Asia weapon).** China controls **~98% of rare-earth processing** and wields **export controls** (gallium, germanium, antimony, tungsten, graphite, rare earths) as a coercive weapon — currently in a *temporary one-year suspension* (Nov 2025–Nov 2026) but with **targeted use continuing** (June 2026: 10 US firms restricted; July 2026: dual-use blocks on 14 EU firms). This is the **commodity mirror of the financial freeze** — see the cross-cutting section.

South America — oil, lithium, and a live intervention

- **Venezuela (it happened).** The "US-invades-a-neighbor" case went live in 2026: a **US intervention, Maduro captured**, and Venezuela's **Essequibo** claim on Guyana put on ice. Instrument: **oil; PDVSA/ Venezuela bonds** (a regime-change / sanctions-relief *recovery* trade); **ExxonMobil / Guyana** (Exxon pumps ~900k bbl/d from the disputed offshore). A *localized* war the great indices ignored — priced entirely in the oil-and-specific-credit complex, exactly as the framework says.

- **Guyana.** From frontier to one of the world's fastest-growing oil provinces; the Essequibo dispute was the territorial-oil overlay. Instrument: **Exxon, Guyana sovereign/oil**.
- **Argentina.** The Milei reform-or-relapse; instrument: **Argentine bonds, the peso, YPF, Vaca Muerta shale, lithium** — an EM turnaround *allocation*, not an event.
- **The lithium triangle** (Argentina/Chile/Bolivia) and **Brazil** (Petrobras, iron ore, soy) — critical-mineral and commodity instruments; localized politics priced in the *specific* names.
- **Mexico.** The nearshoring beneficiary *and* the tariff target *and* the security story — a three-way. Instrument: **MXN and Mexican equities**, acutely sensitive to **US tariff announcements** (trade the print).

Africa — the mineral supply, contested

- **DRC** — **cobalt**, the battery chokepoint (M23/Rwanda conflict over mineral-rich east). Instrument: cobalt and battery-supply names.
- **Sahel (Niger/Mali/Burkina)** — coups, Russia/Wagner, and **uranium** (Niger supplied French reactors). Instrument: uranium and the exposed utilities/miners.
- **Sudan** (gold, civil war); **Nile / GERD** (Egypt–Ethiopia water). Localized, commodity-specific.

The Americas' chokepoint — Panama

- **Panama Canal (live).** A *double* chokepoint: **drought** (Gatun Lake levels, El Niño risk → auction-slot suspensions, July 2026) *and* **geopolitics** (US–China pressure over the ports; the port-concession ruling, Jan 2026). Instrument: **shipping/freight, canal transit, port operators** — the Western-hemisphere Red Sea.

Two cross-cutting instrument-families

Widening the map, the same two families keep carrying the risk — and neither is the equity index.

1. Chokepoints. Tiran (1967) → Suez → Hormuz (Tanker War; Iran) → **Bab-el-Mandeb (Red Sea)** → **Panama**. A chokepoint prices in **freight and tanker rates, oil/LNG routing, and war-risk insurance**, and its resolution is binary and datable (open or shut). Today there are **two live at once** (Red Sea, Panama) plus the Hormuz overhang — a rich, recurring, tradeable family.

2. Critical minerals — the chokepoint applied to supply chains, and a weapon. Rare earths (China ~98% of processing), gallium/germanium, cobalt (DRC), lithium (South America), uranium (Niger). China's export controls are the **commodity mirror of the 2022 financial freeze** — a state-wielded coercive weapon. And it is the *same wasting asset*: the effect is real, the *decisiveness* is contested (hence the 2026 suspension), and the West is running the **"Never Again" engine** — stockpiling, re-shoring processing (MP Materials, Lynas), building alternative supply — which erodes the weapon's future bite exactly as China's gold erodes the dollar-freeze's. Instrument: **the miners/processors and the exposed defense/AI/EV supply chains** — never the index.

The grid, restated for the whole map

ASK	SORTS EVERY REGION
Could it go general?	Taiwan/SCS (candidate general → un-priceable); Iran/Red Sea/Panama/Venezuela (localized → priced in the *exposed* instrument); NK/India-Pak (localized, tail)
Which instrument?	oil, chips, **memory**, **freight rates**, **rare earths/cobalt/lithium/uranium**, PDVSA/Argentine bonds, MXN/EGP, defense equities — **never the index**

The throughline, 2026: two live chokepoints (Red Sea, Panama) and a Hormuz overhang; a critical-minerals weapon that mirrors the financial freeze and is decaying the same way; a US intervention in Venezuela that priced only in oil and specific credit; and — across all of it — calm headline indices that are, almost everywhere, the wrong instrument.

Sources (cited, contemporary, [VERIFY]): CNBC/CSIS on Venezuela–Guyana; Fastmarkets/INN on China's rare-earth controls; Panama Canal Authority / TradeWinds on the canal; Container-News/S&P on the Red Sea. Framework and risk only; not investment advice.

PART VII

Reading the Forward Curve

The Ukraine lesson generalizes: for any conflict, the market's forward expectation — how long it thinks the disruption lasts and where it settles — lives in the term structure of the exposed instrument, not the spot price. But the shape *that term structure takes changes with the instrument. This note does oil/Iran (the direct analog), then the general law, then the one unifying pattern that ties it to the discrimination ladder.*

Oil / Iran — the direct analog to Ukraine gas

Brent's forward curve carries the same forward information the TTF curve did for Ukraine. Crude is normally in **backwardation** (front above deferred). When Hormuz is threatened:

- the **front spikes and backwardation steepens** — the market pricing *acute near-term* supply risk it expects to *ease* (the classic "temporary" signal);
- the **prompt-vs-deferred spread** is the tradeable read of **"how long does the market think the strait stays contested?"** — a wide front premium says *weeks*, a lift in the whole curve says *structural*;
- **June 2025** (scare, strait open): the front jumped and *fell back below* start within days — the curve said "temporary," and it was; **spring 2026** (strait actually shut): the front ran to $114 and **held above $100 for two months** — the curve stayed elevated further out, pricing a *real*, *persistent* disruption. Same instrument, and the *duration* of the backwardation discriminated scare from closure.

So the oil read is identical in structure to the wheat/gas read: **backwardation depth = how temporary; the out-months = where it settles; the calendar spread = the timing of resolution; options skew = the escalation tail.** (Spot arc, Brent: ~$75 pre-war → ~$108 2022 → ~$83 Aug 2026 — reproducible `BZ=F`; the live signal is in the contract-month spreads.)

The general law — every instrument has its term structure

The Ukraine/oil insight is a special case of a universal one:

> ***The spot price says the crisis is here. The term structure of the exposed instrument says what the market expects next — its duration and its resolution.***

What plays the role of "the curve" depends on which instrument carries the risk:

CONFLICT TYPE	EXPOSED INSTRUMENT	ITS "TERM STRUCTURE"	AN *INVERSION* MEANS
Commodity (Iran oil, Ukraine gas/wheat, Red Sea)	oil, gas, wheat, freight	**futures backwardation / calendar spreads**; FFAs for freight	acute-now, expected to *ease*
Sovereign / credit (Russia, Venezuela, Argentina, a default)	CDS, sovereign bonds	**CDS term structure & the bond curve** — short-dated CDS *above* long	a near-term event (default/survival) the market expects to *resolve soon*
Currency (Brexit, EM pegs, the ruble)	FX	**forward points / NDFs & risk-reversal skew**	expected *devaluation* (and the NDF reveals what a *managed* spot hides)
Equity / systemic (Taiwan, general-war scares)	index / sector vol	**VIX term structure & options skew** — spot vol *above* forward vol	acute fear expected to *pass*
Rates / fiscal (US debt ceiling)	Treasury bills	**the bill maturity ladder** — a *kink* at the X-date	the market dating *where* the risk concentrates
The decision itself (any)	prediction markets	**the maturity ladder of odds** (P by Dec vs by Jun)	the crowd's *timing* of the political decision

The move is always the same: **don't read the spot — read the shape.** An inverted/backwardated structure is the market saying *"severe now, survivable, temporary."* A flattening or a lifting of the back end is it saying *"this is becoming structural."*

The one unifying pattern (and the tie to the ladder)

Across all of them, the term structure behaves the same way by rung:

- **Localized / survivable disruption → the curve *inverts* (backwardation, inverted CDS, inverted VIX):** the market prices an acute-but-temporary event and tells you, in the shape, *when* it expects resolution. This is where the forward information is richest and most tradeable (Ukraine gas, Iran oil, a sovereign in an event).
- **Existential / un-priceable → the curve goes *silent*:** for a true Taiwan-style war, the VIX term structure does *not* invert dramatically and the far-dated tail stays cheap — **because you cannot price a catastrophe you will not survive to collect on.** The *absence* of a term-structure signal is itself the tell that you've hit the ceiling. A flat curve into a genuine existential risk is not calm — it's un-priceability.

So the term structure is the same tool everywhere, and it even diagnoses its own limit: **it carries forward information precisely where the risk is survivable, and it falls silent exactly where it isn't.**

The discipline (unchanged)

The curve prices the **exposed channel's** expected resolution — supply for commodities, default-timing for credit, devaluation for FX — **not the underlying military/political decision.** It is the anticipation channel operating on the *handicappable sub-question* ("how long is Black Sea grain blocked," "how long is Hormuz contested," "does this sovereign survive the next coupon"), which it does well, and it is *not* a forecast of "will they escalate," which it does badly. Read it as *expected disruption duration*, a proxy for the conflict's intensity — never as a war oracle. And where you want the decision itself, that's what the **prediction-market maturity ladder** is now for — the one term structure that prices the political choice directly (thin and manipulable, so a lead, not a truth).

Worked example with data and chart: `ukraine-futures-forward-info.md` *(TTF gas 12× then normalized; the 2022 backwardation's "temporary" call, realized). Framework/risk only; not advice; the term-structure signal sits in the individual contract months (ICE/CME), beyond the spot series here.*

Ukraine — what wheat and energy futures actually tell you about the war

Short answer: yes, there is forward information — but it lives in the term structure (the shape of the forward curve and the calendar spreads), not the spot price. The curve encodes the market's expected duration and resolution of the supply disruption, and in 2022 it beat the narrative. The discipline: it prices the supply channel's expected normalization — a proxy for the conflict's expected intensity — not the military/political decision itself.

The exposed instruments, and the spot arc

Ukraine's war prices through **wheat** (Russia + Ukraine ≈ a third of world wheat exports; Black Sea ports blockaded) and **European gas / TTF** (Russian pipeline flows cut). The spot record:

- **Wheat (Chicago, ¢/bu):** 671 (Jun '21) → **1006 on the invasion (+50%)** → 809 (Aug '22) → 665 (Jul '23) → 559 (Jan '25) → **538 (Jan '26)** → 640 (Aug '26). Spiked, then **normalized below the pre-war level** as global supply adjusted and Ukraine re-routed via the Danube and land.
- **TTF gas (€/MWh):** 35 (Jun '21) → 126 (Mar '22) → **240 (Aug '22, ~7× — intramonth higher)** → 28 (Jul '23) → 53 (Jan '25) → 39 (Jan '26) → 55 (Aug '26). Spiked to a generational extreme, then **collapsed back to ~€30** as Europe re-balanced (LNG, demand destruction, storage) — settling at a structurally-higher "new normal" (~€30–55) than the pre-war €20.

Chart: `war_premia/results/ukraine_wheat_gas.svg`.

Where the forward information actually is — the term structure

Spot tells you the crisis is *here*. The **curve** tells you what the market expects *next*:

- **Depth of backwardation (front months above deferred) = how *temporary* the market thinks it is.** At the 2022 peaks, both wheat and TTF were **steeply backwardated** — the front bid far above the out-years. That is the market saying, in price, *"acute now, but we expect it to ease."* **It was right** — the deferred contracts were pricing the normalization that arrived in 2023–24, while commentators were forecasting a permanent energy crisis.
- **The out-year forward level = where the market thinks it *settles*.** TTF forwards for 2024–25 sat far below the €240 front — encoding "Europe solves this, but at a higher floor than pre-war." Also realized (~€30–55).
- **Calendar spreads = the market's *timing* of resolution.** The front-vs-deferred spread is the tradeable "when does the disruption clear?" And for gas specifically, the **winter-vs-summer spread** is the *war-risk premium priced into the heating season* — the cleanest single read of expected war-related tightness.
- **Options skew / implied vol = the *escalation tail*** — how much the market pays for a further supply shock.

It updates on the war's events (the resolution channel, live)

The curve reprices cleanly on each **supply-relevant** decision — which is exactly the "trade the resolution" pattern: - **Black Sea Grain Initiative (Jul 2022)** → wheat backwardation collapsed (exports resumed). - **Russia exits the deal (Jul 2023)** → re-steepened, but *less* — the market had learned the alternative routes worked (adaptation priced in). - **Nord Stream cut / sabotage, pipeline and port attacks** → TTF front spikes, winter spread widens.

So you can read the market's evolving expectation of the war's *supply impact* off the curve in real time.

The discipline — what it does and does not forecast

This is the framework's anticipation/resolution line, applied: - The curve is **good** at the *supply-logistics* question — "how long will Black Sea grain be disrupted," "can Europe replace Russian gas" — because that is a handicappable, physical problem. There it carried genuine forward information and **beat the narrative**. - It is **not** a forecast of the *military/political* decision — "will Russia escalate, will there be a ceasefire." Those are the un-handicappable decisions the anticipation channel is weak on. - So the forward info is a **proxy for expected disruption duration/intensity**, not a war oracle. Read it as "the market expects the *supply* effect to last X and settle at Y," not "the market thinks the war ends in Q3."

What the curves say *today* (Aug 2026)

Both have **ticked up** off their early-2026 lows — wheat ~640 (Kansas ~714), TTF ~€55 (from €39 in January). The framework's instruction: **don't read the spot tick — read the spread.** - If the move is **front-loaded backwardation**, it's a *near-term* supply issue (weather — the Kansas premium hints at US drought) that the curve expects to pass. - If the **whole curve lifts** (out-years up, winter spread widening), the market is pricing a *persistent* war-related tightening into the heating season.

That distinction — weather vs war — is exactly the forward information you can extract right now, and it's in the calendar/winter spreads, not the headline price.

The tradeable version

The calendar spread *is* the "when does the market think this resolves?" trade: own front-vs-deferred to express a view on disruption duration; the **winter-vs-summer TTF spread** to trade the war-risk premium in the heating season; options skew for the escalation tail. And watch them **update on the grain-corridor and pipeline events** — that's where the resolution repricing lands.

Data: Yahoo `TTF=F` (Dutch gas), `ZW=F` / `KE=F` (wheat), `BZ=F` (Brent), reproducible. Term-structure / calendar-spread detail requires the individual contract months (ICE/CME) — the spot series here shows the arc; the forward curve is where the live signal sits. Framework/risk only; not advice.

PART VIII

Finding & Trading a Mispricing

"Mispricing" here has a precise meaning: not "the market is dumb," but the decision-odds (from prediction markets) diverge from the disruption premium the exposed instrument is actually pricing. The framework gives you a repeatable detector — decompose, compare, trade the gap. Illustrative of the method; live inputs (Kalshi P, oil implied-vol/skew, the freight curve) come from a data feed — this book's stock-picker already carries the Kalshi side.

The detector (the recipe)

1. Find the **exposed instrument** (oil for Iran, freight for the Red Sea, semis for Taiwan — never the index).
2. Read its **implied premium** — implied vol / skew / term-structure backwardation — i.e. the probability-weighted disruption the *instrument* is pricing.
3. Read the **decision-odds** for the same event off the **prediction market** (Kalshi/Polymarket P).
4. The **mispricing is the gap**: `P(prediction market)` vs `P implied by the instrument`. - **Odds rich, premium cheap** → the instrument is *under-pricing* the tail → **buy convexity**. - **Premium rich, odds cheap** → the instrument is *over-pricing* it (a fading scare) → **sell/fade**.
5. Trade the gap with **defined-risk convexity in the exposed instrument**; size as insurance; the kill is the gap closing against you.

The key move: `oil premium ≈ P(disruption) × size`. Read **P** off Kalshi, **size** off the oil curve — and if the two Ps disagree, one of them is the trade.

The worked example — Hormuz, and why it's two-sided *right now*

The instrument read (live). Oil implied vol (OVX) is **~55** as of early August 2026 — *elevated*, off a mid-July spike to ~64, against a calm-regime ~27–30 and a spring-2026 crisis peak of ~108. So oil is **already pricing meaningful Hormuz/Red Sea risk** — it is *not* complacent today.

That changes which side of the trade is live:

- **If Kalshi's P(Hormuz closure/incident by year-end) is *below* what OVX ~55 implies** → oil vol is **rich**, the market is over-paying for a tail the odds don't support → **fade it**: sell the rich front vol / upside as a *spread* (never naked), or sell a call-spread against a real-closure cap. This is the *scare-premium harvest* from the "trading the quiet" playbook — and with OVX elevated, this is the more likely live side.
- **If Kalshi's P sits *above* the OVX-implied premium** → oil is under-pricing the tail → **own Brent call spreads / a call calendar** (buy the cheap deferred, sell the collapsed front).

Why the calm version is the cleaner recurring trade. The framework predicts a *systematic* version of the "buy" side: **when a scare resolves and OVX mean-reverts to ~30, oil forgets the standing risk (it's a resolution-pricer), while Kalshi keeps a standing decision-probability.** That is the recurring mispricing — *cheap tail in the calm* — and the disciplined rule is: **fire the buy when OVX is back near its calm floor and Kalshi P is still non-trivial; fire the fade when OVX is elevated and P is lower than the premium implies.** Today (OVX ~55) the window for the buy is *closed*; the fade is the candidate — pending the actual Kalshi print from the feed.

The trade, either way: defined-risk (spreads, calendars), insurance-sized, and it pays/loses on the *resolution* — a real Hormuz incident closes the gap violently in the buyer's favour (front vol and backwardation re-spike, as in spring 2026); a durable ceasefire closes it in the fader's favour.

A second, different *type* — a duration mispricing (Red Sea freight)

Different curve, different failure mode. Shipping **forward freight (FFAs)** may price a **quick Red Sea reopening** (near-term rates high, forwards sloping to normal) while the framework's read *and* the shipping consensus (Houthis persistent, Cape routing the default **through 2027**) say the disruption is **structural**. If the curve prices *temporary* where the reality is *persistent*, **deferred freight is too cheap** → own deferred FFAs / the shipping equities the curve is pricing to normalize but won't. Mirror image if a reopening is nearer than the curve thinks. The gap here is **duration**, not probability — same recipe, different instrument.

From trade to *sleeve* — the systematic version

The same detector becomes a stock-selection tilt, which is why it belongs as a sleeve in the equity book (see `pari_mutuel_geopolitical_sleeve.md`). For each live event with a Kalshi probability:

> ***signal(stock) = P(event | Kalshi) × exposure(stock → the event's instrument) − premium already in that instrument (OVX / skew / curve)***

Go **long** the exposed equities when the odds are rich and the instrument premium is cheap (the market hasn't put the decision-odds into the names yet); **stand down / trim** when the premium already reflects the odds. Examples of the exposure map: Iran/Hormuz → energy & tanker names; Red Sea → shipping & freight; Taiwan → semis (insurance only — un-priceable); rare earths → miners/processors; rearmament → defense primes; Venezuela → oil & specific E&P/credit. The sleeve is long-only and tilts the existing universe; it does not try to time the war — it tilts toward names carrying a **decision-odds-vs-premium gap** the rest of the book isn't pricing.

The honest caveats

- **This is the detector, not a live call** — the exact gap needs the live Kalshi P and the oil vol/skew/freight curve; OVX ~55 today says the *buy-the-calm* side isn't open, which is itself the discipline (don't buy a tail that's already priced).
- **Right on instrument, hard on timing** — defined-risk, convex, patient; the edge is the *detection* (the framework tells you where the gap can be), not the when.
- **Prediction-market P is thin and manipulable** — a lead, not a truth; verify the print before trusting the gap.

Companion to `trade-expressions.md`, `forward-curve-across-conflicts.md`, and the geopolitical sleeve. Framework and risk only; not investment advice.

The geopolitical mispricing sleeve (pari_mutuel_trader)

A new sleeve in the weekly long-only US-equities stock-picker. It turns prediction-market decision-odds (Kalshi/Polymarket P) into a per-name equity tilt via the instrument-problem logic: the risk of a conflict lives in a specific exposed instrument, and the edge is the gap between the decision-odds and the premium the instrument already prices. It plugs in as an ordinary `Agent`, so it is softmax-pooled, hedge-weighted, and separately attributed like every other book.

The signal

For each live event `e` with market probability `prob_e` (from the Kalshi feed) and an implied premium `premium_e` already in its instrument (both in [0, 1]):

`geo_signal[symbol] = Σ_e  (prob_e − premium_e) · exposure[e][symbol]`

- **`exposure` is signed.** A name that *benefits* from the disruption (an oil major when Hormuz is threatened; a defense prime on rearmament; a rare-earth miner on a China squeeze) has a **positive** weight; a name that is *hurt* (a chip designer if Taiwan is threatened) has a **negative** weight.
- **`edge = prob − premium`** is the mispricing. **Positive edge** (odds richer than the premium) → tilt **toward beneficiaries, away from victims** (the market hasn't put the odds into the names yet). **Negative edge** (the premium is already rich — a fading scare) → the reverse (fade / stand down).
- With no events supplied, the sleeve is **neutral** (all-zero) — it degrades gracefully, exactly like the news/macro agents when their column is absent.

The default exposure map (`data/geopolitical.py: DEFAULT_EXPOSURE_MAP`) covers **Iran/Hormuz** (energy & tankers up, airlines down), **Red Sea** (shipping up), **Taiwan** (semis down — un-priceable, trim only), **rare earths** (Western pure-plays up), **rearmament** (defense up), and **Venezuela** (oil/E&P). Override it for a live book.

Instrument notes — which proxy actually carries the trade

The exposure map is only as good as the instrument behind each name. Two cases sit on opposite sides of the instrument problem:

- **Rare earths — no clean instrument exists.** There is *no liquid Western-listed rare-earth (NdPr) future*; the metal prices on Chinese/OTC venues. So the theme is **forced into equity proxies**: **MP** is the only scaled US pure-play (magnet-metal miner); **LYSCF** (Lynas) is the ex-China pure-play but trades US OTC; **REMX**, the obvious ETF, is **contaminated** — it holds Chinese producers and lithium and can move the *wrong way* in a squeeze, so it carries only a small weight. A prior version mislabeled **ALB** (lithium) and **UEC** (uranium) as rare earths; they now sit in their own `LITHIUM` / `URANIUM` themes so they don't distort the rare-earth signal.
- **Oil — many instruments, ranked by contamination.** The *premium* prices in crude futures and **OVX**; use **Brent** (the Hormuz-relevant waterborne barrel), not landlocked WTI. **USO** bleeds roll yield in contango — never a hold; **BNO** (Brent) is the cleaner ETF. For a long-only stock book the expression is integrated/E&P equity plus **tankers** (FRO/STNG/INSW/DHT), which often lead on closure/reroute *fear* even when barrels are ultimately rerouted and flat price gives the move back.

How it plugs in (no engine changes)

Following the repo's conventions exactly: - `agents/geopolitical.py` — `GeopoliticalAgent(name="geopolitical")` reads the `geo_signal` column (zero-fill if absent), like `NewsIntensityAgent`. - `agents/__init__.py` — registered in `build_v1_agents()`; the engine softmax-pools it into the pari-mutuel book, gives it a hedge-learned weight (starts 1/N, then `hedge_update` rewards it on realized performance), and attributes it separately. - `data/features.py` — `geo_signal` added to the feature whitelist (default 0.0). - `data/geopolitical.py` — `build_geo_signal()`, `attach_geo_signal(features_df, events)`, `load_events(path)`.

Feeding it live — Kalshi is wired into `build-features`

The sleeve is populated **automatically** each time you build features. `cli.py: build-features` reads `data.geopolitical_path` (default `configs/geopolitical.yaml`), resolves each event's `prob` from Kalshi, and attaches the `geo_signal` column — no manual step. It prints, e.g. `geo sleeve: attached geo_signal for 5 events (prob source: ['kalshi_live', 'static'])`.

Two inputs per event: 1. **`prob`** — the decision-odds, resolved in priority order by `data/kalshi.py`: **live** (GET the Kalshi market named by the event's `kalshi_ticker`) → **local file** (`KALSHI_PROBS_PATH`, a `{ticker: prob}` CSV/JSON your pipeline writes) → **static** (the `prob` in the config). So it works with no keys/network (static) and upgrades to live odds when a feed exists. 2. **`premium`** — the disruption already priced in the exposed instrument, now **auto-fetched from implied vol** (`data/premium.py`): **OVX** for oil events, **MOVE** for rate/macro events, **VIX** for equity/systemic events — normalized to [0,1] as the fraction of the instrument's crisis range currently implied. Events with no clean vol proxy (rare earths, rearm, FDA) keep the static config `premium`. This is what makes it a *mispricing* signal, not a fear gauge.

Config: copy `configs/geopolitical.example.yaml` → `configs/geopolitical.yaml`, give each event a `kalshi_ticker` (omit it for structural themes like `REARM` to keep them static), and set env in `.env` (`KALSHI_API_BASE`, `KALSHI_API_KEY`, `KALSHI_PROBS_PATH` — all optional). Then just `build-features` as usual and the tilt is live.

Tying it to the existing sleeves — the Kalshi→macro_regime bridge

The sleeve is not a bolt-on; `edge = prob − premium` **is** the pari-mutuel objective the whole engine runs on (true odds − pool-implied odds), just read across two venues (Kalshi vs the vol market). To make that tie explicit, the macro events feed the repo's existing **`MacroRegimeAgent`** directly:

- `data/geopolitical.py: build_macro_regime(events)` projects the FED/CPI/RECESSION edges onto one risk-on/off axis — `regime = clip(Σ (prob − premium)·sign, −1, 1)`, with `FED_CUT +1`, `FED_HIKE / CPI_HOT / RECESSION −1`. A **cheap premium** (calm MOVE/VIX today) with standing odds → a large-magnitude regime; an **already-rich premium** → near zero — the same discipline as the name tilt.
- `attach_macro_regime(features_df, events)` adds that scalar into the `macro_regime` column (additive + clipped, so it composes with any returns-derived regime). `MacroRegimeAgent` then turns it into a per-name tilt via its existing `regime · (ret_20d − vol_20d)` — **prediction-market odds now set the regime sign** instead of (or on top of) realized returns.
- `build-features` wires it automatically and prints, e.g., `geo->macro bridge: macro_regime -0.42 from Kalshi macro odds`.

So one Kalshi read now drives **two** sleeves — the geo name-tilt *and* the macro regime — expressing the same view through two lenses. Any redundancy is handled by the hedge learner, which sizes each sleeve on realized P&L. (Companion options not yet wired: an `edge = signal − priced` house primitive retrofit to momentum/news, and geo-sign gating of momentum entries — "trade the resolution.")

The discipline (built into the design)

- **Two-sided, and honest about the regime.** With **OVX ~55 (Aug 2026, elevated)**, the oil premium is high, so `IRAN_HORMUZ` edge is likely ~0 or negative right now — the sleeve *tilts nothing* (or fades), which is correct: don't chase a tail already in the price. The clean *buy* fires when OVX mean-reverts to its calm floor (~30) while Kalshi keeps a standing probability.
- **Un-priceable ≠ tradeable.** Taiwan carries *negative* exposure (trim chip names) only — you can't harvest a premium the market can't price; the sleeve treats it as light insurance, never a short.
- **Prediction-market P is thin and manipulable** — a lead, not a truth. It enters as one input among the pari-mutuel book's several agents, and its influence is *hedge-weighted down* automatically if it doesn't perform. That is the right home for a noisy-but-informative signal: co-equal, attributed, and self-correcting.

What the recommendations actually look like

The sleeve's output is a **per-name tilt** (`geo_signal`), not a buy list — the engine pools it with the other agents and the hedge weight decides how much it moves the book. But read on its own, with the example config (Aug 2026 odds; premiums auto-fetched), the tilts come out like this:

EVENT	EDGE = PROB − PREMIUM	NAMES IT TILTS (WEIGHT = EDGE × EXPOSURE)
REARM (structural, static premium)	0.70 − 0.40 = **+0.30**	**LMT / RTX / NOC +0.30**, GD / LHX +0.24, HII +0.18
RED_SEA	0.55 − 0.35 = **+0.20**	**ZIM +0.24**, MATX +0.20, XOM +0.08, CVX +0.06
RARE_EARTH (static premium)	0.40 − 0.20 = **+0.20**	**MP +0.30**, ALB +0.12, UEC +0.12
TAIWAN (un-priceable → trim only)	0.08 − 0.03 = **+0.05**	**TSM −0.06**, NVDA / AMD −0.05, AVGO / ASML / MU −0.04, QCOM −0.03
IRAN_HORMUZ (OVX elevated)	~0.30 − ~0.30 ≈ **0**	**no tilt** — the tail is already in the oil price; the sleeve stands down

Read plainly: **long the beneficiaries with a standing odds-vs-premium gap** (defense primes on rearmament, box-shippers on the Red Sea reroute, the one Western rare-earth name), **lightly trim the un-priceable victim** (chip names on Taiwan — insurance, never a short), and **do nothing on Hormuz** because the premium already equals the odds. The single most important line is the last one: the sleeve's job is as much *where not to tilt* as where to.

These are magnitudes on the sleeve's own scale; the book-level position is this tilt × the sleeve's hedge weight, softmax-pooled with every other agent — so no single name is ever a large gross bet.

What a historical return would look like

There are two honest ways to answer "what would this have returned," and they answer different questions:

1. **A strategy backtest** — the engine metric — needs a *dated weekly (prob, premium) panel per event* to replay. We don't have that: Kalshi's markets for these bespoke events are recent and sparse, so a true Sharpe would be fit on a handful of weeks and would overstate. On the synthetic test universe (no real tickers) the sleeve is **neutral by construction** — the engine backtest returns all-zeros / `insufficient_holdings`, which is the correct "no signal on names it can't see." *We do not report a fabricated Sharpe.*
2. **An event study of the tilts** — what the beneficiary basket the sleeve points at actually did, window by window, on **real Yahoo prices** — is defensible as a *documented-effect illustration* (not a live-tradeable P&L, since it uses the realized window with hindsight on dates):

- **Ukraine invasion** (24 Feb → 30 Dec 2022), long the REARM+energy beneficiaries XLE / ITA / LMT / XOM: **+25.8% basket** vs **−10.2% S&P → spread ≈ +36 pts**. (XLE +29.5%, LMT +23.1%, XOM +44.1%, ITA +6.7%.)
- **Red Sea disruption** (15 Dec 2023 → 31 May 2024), long the shipping beneficiaries ZIM / MATX: **+73.3% basket** vs **+10.9% S&P → spread ≈ +62 pts**. (ZIM +124.9%, MATX +21.7%.)

That is what the *exposure map* captured when the resolution channel fired — the strong channel of the whole framework ("trade the resolution, not the forecast"). It is **not** a claim the sleeve would have timed the entries: the discipline is that you only take the beneficiary tilt while `edge = prob − premium` is positive, and you stand down (Hormuz today) when the premium already equals the odds. The event study shows the *size of the move available* when the tilt is right; the `edge` gate is what's meant to keep you from paying for it after it's priced.

Discover more things to price

```
python3 -m pari_mutuel_trader.cli discover --category Economics
```

Enumerates what Kalshi is currently pricing, tags each event that maps to a real exposed instrument (via the keyword→theme map), and prints rows ready to paste into `configs/geopolitical.yaml`. Events with no tradeable instrument (Sports, Pope) or an un-priceable tail (Mars-by-2050) are counted and skipped — the filter from `docs/mining-kalshi-for-instruments.md`, made executable.

Run

```
cd pari_mutuel_trader && export PYTHONPATH=src
python3 -m pytest tests/test_geopolitical.py tests/test_kalshi.py tests/test_premium.py -q
python3 -m pari_mutuel_trader.cli discover --category Economics   # find more instruments
# then feed events via configs/geopolitical.yaml and rebuild features before backtest/paper.
```

Companion: `mispricing-worked-example.md` *(the detector and the Hormuz worked example),* `forward-curve-across-conflicts.md`*,* `live-board-2026.md`*. Framework/risk only; not investment advice.*

PART IX

Mining Prediction Markets for Instruments

Kalshi is a giant map of decision-probabilities. The instrument-problem framework turns each one into a tuple — `Kalshi P(event)` *→ exposed instrument → tradeable names — so "what Kalshi prices" becomes a discovery engine for* more things to price. *The work is the filter: most Kalshi markets are not tradeable this way, and the framework tells you which to keep.*

What Kalshi actually prices (1,200 events, live)

Enumerating the Kalshi event universe (`kalshi.list_events()`), 14 categories:

CATEGORY	~COUNT	FRAMEWORK-TRADEABLE?
Elections	664	**Mostly no** — instrument-less (NATO SG, next Pope). A few map (a G7 leader change → that country's assets).
Politics	178	**Some** — legislation → policy instruments (tariffs, drug pricing, energy).
Entertainment	88	**No** — next Bond, next Rambo. No instrument.
Sports	84	**No.**
Economics	64	**Yes** — Fed funds, CPI, GDP → rates, bonds, banks, cyclicals, gold.
Financials	44	**Partly** — IPO races → the fintech/underwriter complex, comparable publics.
Companies	34	**Yes (single-name)** — antitrust (LYV), CEO succession (JPM), AGI timing → semis.
Science & Tech	21	**Rarely** — fusion → nuclear/uranium; most (Mars-by-2050) are **un-priceable**.
Climate & Weather	11	**Partly** — hurricanes/quakes → insurers, energy, ag; but 2°C-by-2050 / supervolcano are **un-priceable**.
Health	4	**Yes (single-name)** — FDA approvals → the specific biotech.
World / Social	6	**Some** — Trump–Putin meetings, EU expansion → geopolitical.
Crypto	1	**Yes** — BTC/ETH ranges, ETF → COIN, MSTR, miners.

The filter — three tests to keep a Kalshi market

Not every priced probability is a tradeable mispricing. Keep a market only if it passes all three:

1. **Real exposed instrument.** Does `P(event)` map to an instrument whose price moves on the event — oil, rates, a biotech, an insurer, a currency? If not (Sports, Entertainment, the Pope), discard.
2. **Tradeable horizon.** Does it resolve on a horizon you can hold and collect on? A "by 2050" market is not a trade; a Fed decision, an FDA date, a hurricane season, a Hormuz-by-year-end is.
3. **Not un-priceable.** Drop the civilizational tails — 2°C-by-2050, a supervolcano, "colonize Mars." The market cannot price a catastrophe it won't survive to collect on; its Kalshi odds are opinion, not a hedge, and the exposed instrument stays silent (the Cuba ceiling, generalized).

What survives is a **broad, multi-category event universe**, far beyond the six geopolitical events the sleeve started with.

The instrument map (what survives, by category)

KALSHI CATEGORY	EVENT	EXPOSED INSTRUMENT → NAMES (SIGNED)
Macro	Fed hike / higher-for-longer	banks **+** (KRE, JPM), long duration/growth **−** (TLT, ARKK), gold **−**
	Fed cut	long duration **+** (TLT), growth **+**, gold **+**, banks **−**
	CPI hot	energy/materials **+** (XLE, FCX), gold **+**, long bonds **−**
	Recession	defensives **+** (XLP, XLU), cyclicals/HY **−** (XLY, XLI, HYG)
Policy	New tariffs	import-substitutes **+** (X, NUE), importers/targets **−** (GM, NKE, FXI, EWW)
	Drug-pricing reform	pharma/PBMs **−** (PFE, MRK, LLY, CVS, UNH)
Company	Antitrust ruling	the named single-name **−** (e.g. LYV)
	AGI milestone	semis / AI complex **+**
Health	FDA approval	the specific biotech **+**, competitors **−**
Climate	Major hurricane	insurers/reinsurers **−/+** (ALL, TRV; RNR pricing-power **+**), gulf energy/refiners **+**
Crypto	BTC/ETH rally / ETF	crypto-exposed equities **+** (COIN, MSTR, MARA)
Geopolitics	Hormuz / Red Sea / Taiwan / rare earths / rearm	oil, freight, semis (−, un-priceable), miners, defense

These are wired into `DEFAULT_EXPOSURE_MAP` (`data/geopolitical.py`) — geopolitics is now just one block. The sleeve is, in truth, a **prediction-market mispricing engine**; geopolitics was the first category, not the boundary.

The discovery workflow

```
from pari_mutuel_trader.data.kalshi import list_events
# 1. enumerate what Kalshi prices, by category
for e in list_events(category="Economics"):
    print(e["event_ticker"], "-", e["title"])
# 2. keep the ones with a real instrument + tradeable horizon (drop un-priceable/long-dated)
# 3. add {event, kalshi_ticker, premium} to configs/geopolitical.yaml and a row to DEFAULT_EXPOSURE_MAP
# 4. build-features -> the sleeve prices the odds-vs-premium gap across the whole universe
```

Each kept market becomes the same trade the sleeve already runs: **`edge = P(Kalshi) − premium(instrument)`**, tilt toward beneficiaries and away from victims when the odds diverge from what the instrument prices.

Why this matters

The single instrument (oil, semis, freight) was never the point — it was the *first* instrument. Kalshi prices the anticipation channel across **macro, policy, single-name, health, climate, and crypto**, and the framework maps each survivor to its exposed instrument. So "more things to price" is literal: the mispricing detector — *decision-odds vs the premium already in the instrument* — runs on **dozens** of Kalshi markets, not six. The discipline is unchanged and doubly important at scale: keep only the markets with a real instrument and a survivable, near-dated resolution; the rest is noise the crowd is happy to bet on and the market cannot hedge.

Companion: `pari_mutuel_geopolitical_sleeve.md`, `mispricing-worked-example.md`, `forward-curve-across-conflicts.md`. *Framework/risk only; not investment advice; exposure maps are illustrative defaults — verify tickers and signs for a live book.*

PART X

Does the Sleeve Add Value? An Honest Backtest

A walk-forward of the geopolitical/macro sleeve inside the pari-mutuel stock-picker, on a real 65-name universe (2018–2026). The result is a clean negative-turned-marginal-positive that quantifies the book's central discipline — trade the resolution window, not the hold — in the engine's own P&L. Reported with its limitations, because the limitations are the point.

The question

The sleeve turns prediction-market odds into a per-name equity tilt (`edge = prob − premium`). It plugs into the engine as one co-equal, hedge-weighted agent. Does it actually improve risk-adjusted returns — and if not, *why not*, and what would fix it?

Method

- **Universe:** 65 real tickers spanning every exposure theme (energy, tankers, shipping, semis, rare-earth/defense, rate-sensitives, insurance, crypto) plus liquid anchors. Daily prices, Yahoo, 2018-01 to 2026-08.
- **Engine:** unchanged — softmax → pari-mutuel pool → hedge-weighted → top-25 long-only, weekly.
- **Configurations** (same engine, same universe; only the sleeve's input changes):

	CONCENTRATE WHEN LIVE?	EXIT AFTER THE CATALYST?
A — baseline (sleeve off)	—	—
B — static tilt (today's book held constant)	no	no
C — dated, hold-through	no	no
C + conviction	**yes**	no
D — exit rule (signal decays)	no	**yes**
D + conviction	**yes**	**yes**

- **Honesty flag up front:** the dated events are placed at the dates their episodes actually happened (Ukraine 2022, Red Sea 2023-24, the 2022-23 hiking cycle, …). That is **hindsight on entry**. So this tests the *exit discipline given a correct entry* and the *engine machinery* — not real-time entry timing. Identifying the resolution live is a separate job, and it is exactly what the Kalshi `prob` feed is for.

Result 1 — the ladder

Full-period backtest, 2018–2026 (baseline Sharpe **0.461**, CAGR 20.4%):

RUN	CONCENTRATE	EXIT	SHARPE	CAGR	VS BASELINE
A — baseline	—	—	0.461	20.36%	—
B — static tilt	no	no	0.453	20.11%	−0.008
C — dated, hold-through	no	no	0.450	19.61%	−0.011
C + conviction	**yes**	no	**0.442**	19.27%	**−0.019 (worst)**
D — exit rule	no	**yes**	0.455	20.00%	−0.006
D + conviction	**yes**	**yes**	**0.463**	**20.47%**	**+0.002 (beats A)**

Read the four dated rows as a 2×2 and the mechanism is unmistakable:

- **Concentrate *without* an exit is the single worst thing you can do** (0.442). You load into the catalyst names at the resolution and then hold them for years — straight into a decade the tech/AI complex dominated and defense/energy/shipping did not. You concentrate the mistake.
- **Exit alone helps but isn't enough** (0.455): you rotate out in time, but never loaded in hard enough to capture the pulse.
- **Exit *and* concentrate is the only config that beats baseline** (0.463): load in at the resolution, ride the repricing, rotate out as it settles. It needed **both halves**.

That is "buy the invasion, sell the phony war," demonstrated in the engine. The sign of the sleeve's contribution flips exactly where the framework says it should.

Result 2 — is the winning config robust, or a lucky knob?

Sweep the exit half-life for the winning config (baseline 0.461):

HALF-LIFE	SHARPE	VS BASELINE	CAGR
4 wk	**0.467**	**+0.006**	20.65%
8 wk	0.465	+0.004	20.56%
13 wk	0.463	+0.002	20.45%
26 wk	0.455	−0.006	19.93%
52 wk	0.454	−0.007	19.91%

Monotonic, not knife-edge. Sharpe declines smoothly as the half-life lengthens; the config beats baseline across the entire short band (4/8/13 wk) and turns negative only past ~a quarter. The sign is governed by one interpretable parameter — *how fast you exit* — not a fragile sweet spot. The a-priori 13-week pick was conservative; the true optimum is faster.

The economic reading is the whole book in one line: **the resolution edge decays with a half-life of weeks.** Hold the catalyst names a quarter and it's gone; hold a year and you are worse than doing nothing.

What this means — and doesn't

Validated: the *mechanism*. Static directional exposure is a drag; concentration without an exit makes it worse; concentration *with* a weeks-scale exit is the only thing that adds value, robustly. This is the "resolution window, not the hold" discipline, proven in-engine on real prices.

Not claimed: - Not real-time entry alpha. Entries were hindsight-placed. The test isolates the exit discipline and the engine, not the ability to call the resolution live (that is the odds feed's job). - **Not a large edge.** Even the best config is +0.006 Sharpe / +30 bps CAGR. Diluted into a 25-name long-only book at a ~1/7 hedge weight, the magnitude is small. The edge is meant to be harvested in a **concentrated satellite**, which this core-overlay backtest structurally cannot express. - **One universe, one regime.** 2018–2026 was a tech-led bull market with only brief war-resolution windows — a hard environment for this sleeve. A value/defensive-led decade with a major resolution could read differently. (The book-level −28% max-drawdown breach is a config artifact of the equal-weight/no-vol-target benchmark, constant across all rows; it does not affect the comparison.)

The takeaway for how to use it

Keep the sleeve as an **attributed, hedge-weighted lead indicator** and a **satellite trade generator** — not a lever you crank inside the core. When you *do* express it, the engine now carries the discipline as a first-class feature: `attach_geo_signal_timeline(..., half_life_weeks=…)` pulses the signal at the resolution and decays it out, so the weekly rebalance performs the exit. Set the half-life to weeks, not quarters.

Reproduce: `scripts/fetch_real_universe.py` *then* `scripts/wfo_geo_sleeve.py` *(add* `--sweep` *for the half-life curve). Framework and risk disciplines only; not investment advice. Backtest figures are illustrative of mechanism, not a live-tradeable track record.*